ENGR 1451/2451: Exploratory Data Science

Polling and Elections: Sampling, Bias, and Uncertainty

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Today: Polling and Elections

Polls use sample data to estimate population preferences.

- Interpret poll percentages, standard errors, and confidence intervals.
- Discuss chance error and bias.
- Learn from the historical polls and elections to discuss different types of bias.
- Compare probability sampling methods and understand the limits of election forecasts.

What Does a Poll's $48\% \pm 3$ Points Mean?

A hypothetical poll of Blue versus Red

Blue: $48\% \pm 3$ points

Approximate 95% confidence interval: **45% to 51%**

Sample proportion $\hat{p}$

The fraction of surveyed people who choose Blue: **observed**.

Population proportion p

The fraction of all target voters who choose Blue: **unknown**.

Question: How can we quantify uncertainty when p is unknown?

One Voter Is a Bernoulli Trial

Code a randomly selected voter as $X_i = 0$ or 1.

Candidate A (Blue)

$$X_i = 1$$

$$P(X_i = 1) = p$$

Candidate B (Red)

$$X_i = 0$$

$$P(X_i = 0) = 1 - p$$

Mean and SD of *one* voter outcome

$$\mu = E(X_i) = p, \qquad \sigma = \sqrt{p(1-p)}.$$

The Poll Percentage Is a Sample Average

Average of n independent voter outcomes

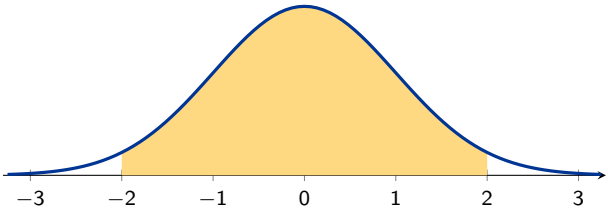
$$\hat{p} = \bar{X} = \frac{X_1 + \cdots + X_n}{n}, \qquad E(\hat{p}) = p.$$

Standard error: variation across repeated polls

$$\mathrm{SE}(\hat{p}) = \frac{\sigma}{\sqrt{n}} = \sqrt{\frac{p(1-p)}{n}}.$$

This is the **same standard-error rule** as for the average of repeated die rolls.

Central Limit Theorem: Why "± 2 SE"?



Large-sample rule

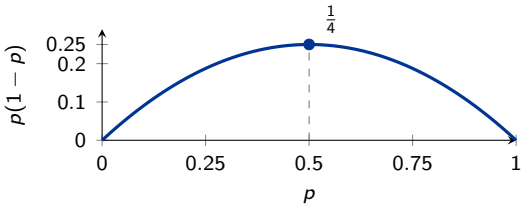
About 95%

within ± 2 SE of p

$$\frac{\hat{p} - p}{\mathrm{SE}(\hat{p})} \approx N(0,1) \quad \text{for a large, suitable random sample.}$$

Unknown p? Use the Largest Possible SE

The term $p(1-p)$ is largest when $p = 0.5$.



Conservative upper bound

$$\mathrm{SE}(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

$$p(1-p) \leq \frac{1}{4}$$

$$\boxed{\mathrm{SE}(\hat{p}) \leq \frac{1}{2\sqrt{n}}}$$

Approximate 95% interval:

$$\hat{p} \pm 2\left(\frac{1}{2\sqrt{n}}\right) = \boxed{\hat{p} \pm \frac{1}{\sqrt{n}}}.$$

Why Polls Often Report ± 3 Points

Hypothetical poll: $n = 1{,}100$

528 of 1,100 respondents choose Blue:

$$\hat{p} = \frac{528}{1100} = 0.48.$$

For an approximate 95% interval:

$$\frac{100}{\sqrt{1100}} \approx \mathbf{3.0} \text{ points.}$$

$$48\% \pm 3 \text{ points}$$

Interval: 45% to 51%.

Conservative $\pm$ amount

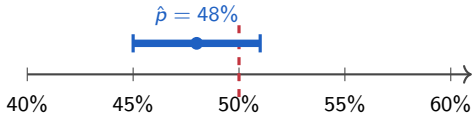
Sample size	$\pm$ points
100	10.0
400	5.0
1,100	3.0
2,500	2.0
10,000	1.0

A Close Poll Does Not Settle the Election

Suppose a hypothetical two-party poll reports

Blue: 48% **Red: 52%**

Blue's approximate 95% interval: **45% to 51%**



What Does “95% Confidence” Actually Mean?

A reported interval for the unknown, fixed population share is 45% to 51%. Which statement is correct?

1. Across many properly conducted polls, about 95% of intervals constructed by a 95% method contain p.
2. There is a 95% chance that p changes within 45% to 51%.
3. We are certain this one interval contains p.

1936: Which Poll Would You Trust?

Two surveys tried to predict the same presidential election.

Literary Digest

2.4 million

returned questionnaires

Gallup

about 50,000

survey respondents

Question: Which poll would you trust more?

1936: The Smaller Poll Was Closer

Roosevelt, major-party vote share	
Actual election result	**62%**
Literary Digest forecast	43%
Gallup election forecast	56%
Gallup's prediction of *Digest*'s result	44%

Surprising result: Gallup used a *separate* sample of about 3,000 to predict the *Digest*'s forecast (44% versus 43%).

Why the *Literary Digest* Poll Failed

1. Selection bias: Who was contacted?

The *Digest* mailed **10 million** surveys using lists such as telephone directories and club memberships.

Many lower-income households were undercovered.

2. Nonresponse bias: Who replied?

Only **2.4 million** questionnaires came back.

$$\text{Response rate} = \frac{2.4}{10} = 24\%.$$

Respondents may differ from nonrespondents.

Key distinction: Being *contacted* and *responding* are two separate selection steps.

More Responses Cannot Eliminate Bias

Two kinds of error

$$\text{estimate} - \text{true value} = \underbrace{\text{bias}}_{\text{systematic}} + \underbrace{\text{chance error}}_{\text{random}}.$$

More observations usually reduce

- random sampling variability

More observations do not fix

- missing groups or selective responses

Perspective: For $n = 2.4$ million, an *ideal random sample* would have a 95% interval of roughly ± 0.06 points, not a 19-point miss.

Dewey Defeats Truman



1948: What Went Wrong With the Polls?

Three major national polls predicted Dewey would win.

	Truman	**Dewey**	**Dewey lead**
Crossley	45%	50%	+5 points
Gallup	44%	50%	+6 points
Roper	38%	53%	+15 points
Actual vote	**50%**	**45%**	−5 **points**

1948: Quota Sampling

How it worked

Each interviewer was assigned fixed **quotas** to fill.

Quotas were based on:

- Residence
- Sex
- Age
- Race
- Economic status

Example: St. Louis

Interview 13 people:

- Residence: 6 suburban, 7 central-city
- Sex: 7 men, 6 women
- Among the 7 men:
  - Age: 3 under 40, 4 aged 40+
  - Race: 1 Black, 6 White
- Similar quotas for women

This seems like a reasonable way to obtain a representative sample.

Why Can Quota Sampling Be Biased?

Three potential problems

1. Interviewer discretion: Interviewers choose whom to approach, often selecting people who are easiest to reach. Human choice is subject to bias.
2. Unmeasured characteristics: Matching age, sex, race, and residence does not guarantee matching political preferences.

Example

An interviewer fills a quota for women under 40 by interviewing shoppers during the day.

Who might be missing?

Three Ways to Select a Probability Sample

Simple random (SRS)

Randomly select voters from the **full list**.

Example: 1,000 names from all voters.

Stratified random

Split into groups, then **randomly select** within *each* group.

Example: sample within urban, suburban, rural.

Multistage / cluster

Randomly select areas, then smaller units.

Example: towns $\rightarrow$ blocks $\rightarrow$ households.

Common feature: Selection is driven by chance, not an interviewer's judgment.

Does a Bigger State Require a Bigger Poll?

Suppose each state randomly samples **2,500 voters** without replacement.

New Mexico (illustrative)	Texas (illustrative)
1.5 million	**15 million**
voters	voters

Questions:

1. Which state has a larger margin of error?
2. Does Texas need 10 times as many respondents to achieve the same precision?



Sampling Without Replacement

Usual SE formula assumes independent observations:

$$\text{SE}(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

But sampling without replacement introduces dependence.

Finite Population Correction (FPC)

$$\text{SE}(\hat{p}) = \sqrt{\frac{p(1-p)}{n}} \underbrace{\sqrt{\frac{N-n}{N-1}}}_{\text{FPC}}$$

If $n = N$: The entire population is sampled, so the sampling error is zero.

Does Population Size Matter?

Conservative 95% margin of error ($n = 2{,}500$):

$$\text{MOE} \approx \frac{100}{\sqrt{2500}} \times \text{FPC} = 2 \times \text{FPC} \quad \text{percentage points}$$

	Population (N)	FPC	MOE
New Mexico	1.5 million	0.99917	± 1.998
Texas	15 million	0.99992	± 2.000

Analogy

Suppose a chemist analyzes a drop of liquid from a well-mixed solution.

Doesn't matter whether the drop comes from a test tube or a gallon jug.

In either case, the drop should reflect the composition of the whole solution.

Beyond Sampling Error

Poll measures current preferences, but election forecast must predict the final vote.

1. Who will actually vote?

Not all eligible voters turn out.

Pollsters must estimate which respondents are **likely to vote**.

2. What about undecided voters?

Some voters have not made up their minds.

Pollsters must estimate how these voters will ultimately vote.

1992 Presidential Election

	Forecast	Actual
Clinton	49%	43.0%
Bush	37%	37.4%
Perot	14%	18.9%

The forecast overestimated Clinton and underestimated Perot.

Beyond Sampling Error

Two different quantities:

- **Ideal 95% MOE:** Random sampling uncertainty at the time of the poll.
- **Forecast miss:** Difference between a predicted and actual candidate vote share.

Candidate (year)	Likely-voter n	Ideal SRS MOE	Forecast miss
Eisenhower (1952)	3,350	± 1.7	**3.9 points**
Clinton (1992)	1,600	± 2.5	**6.1 points**
Bush (2000)	2,400	± 2.0	0.2 points

Example: In 1952, Gallup forecast 51% for Eisenhower; he received 54.9%—a miss of 3.9 points.

How Do Pollsters Address Nonresponse Bias?

Even with random sampling, people who respond may differ from those who do not.

1. Increase response rates

- Contact people multiple times.
- Call at different times of day.
- Use different contact methods.
- Offer incentives.

2. Weight the responses

Suppose younger voters are less likely to respond.

	Population	Sample
Under 40	50%	30%
40+	50%	70%

Give younger respondents **more weight** to restore the 50/50 proportions.

3. Check for remaining bias

Compare results with known population benchmarks. But weighting cannot necessarily correct differences *within* demographic groups.

The Question Can Change the Answer

Historical example: candidate order mattered

In 1948 survey research, placing a candidate's name first changed reported support by about 5 percentage points.

Even if voter preferences do not change, responses can depend on:

- Wording of the question
- Order of response options
- Willingness to respond honestly
- Interview setting or method

Summary: What to Remember About Polls

1. Estimate: The sample percentage $\hat{p}$ is a *statistic* used to estimate the unknown population *parameter* p.
2. Selection: Probability sampling uses chance. Ask who was sampled, how they were selected, and who responded.
3. Bias: Increasing the sample size does not eliminate systematic bias.
4. Chance error: The standard error (SE) measures sampling variability. For an ideal SRS, it decreases roughly as $1/\sqrt{n}$.
5. Confidence: For a large, suitable random sample, about 95% of intervals $\hat{p} \pm 2\ \mathrm{SE}(\hat{p})$ contain the fixed p in repeated sampling.
6. Limits of $\pm$: The margin of error measures chance error, not bias.
   Simple-random-sample SE formulas do not automatically apply to other designs.